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Long-term assessment of nutrient flow pathway dynamics and in-stream fate in a temperate karst agroecosystem watershed

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Abstract

Nutrient dynamics in karst agroecosystems remain poorly understood, in part due to limited long-term nested datasets that can discriminate upland and in-stream processes. We present a 10-year dataset from a karst watershed in the Inner-Bluegrass Region of central Kentucky, consisting of nitrate (nitrate-N [NO_3^-]), dissolved reactive phosphorus (DRP), total organic carbon (TOC), and total ammoniacal-N (TAN) measurements at nested spring and stream sites as well as flowrate at the watershed outlet. Hydrograph separation techniques were coupled with multiple linear regression and Empirical Mode Decomposition time-series analysis to determine significance of seasonal processes and to generate continuous estimates of nutrient pathway loadings. Further, we used model results of benthic algae growth and decomposition dynamics from a nearby watershed to assess if transient storage in algal biomass could explain differences in spring and downstream watershed nutrient loading. Results highlight statistically significant seasonality for all nutrients at stream sites, but only for NO_3^- at springs with longitudinal variability showing significant decreases occurring from spring to stream sites for NO_3^- and DRP, and significant increases for TOC and TAN. Pathway loading analysis highlighted the importance of slow flow pathways to source approximately 70% of DRP and 80% of NO_3^- . Results for in-stream dynamics suggest that benthic autotroph dynamics can explain summer deviations for TOC, TAN, and DRP but not NO_3^- . Regarding upland dynamics, our findings agree well with existing perceptions in karst for N pathways and upland source seasonality but deviate from perceptions that karst conduits are retentive of P, reflecting the limited buffering capacity of the soil profile and conduit sediments in the Inner-Bluegrass. Regarding in-stream fate, our findings highlighted the significance of seasonally driven nutrient processing in the bedrock-controlled streambed to influence nutrient fluxes at the watershed outlet. Contrary to existing perceptions, we found high N attenuation and an unexplained NO_3^- sink in the bedrock stream, leading us to postulate that floating macrophytes facilitate high rates of denitrification.

KEYWORDS

bedrock stream, hydrologic pathways, in-stream fate, karst agroecosystem watershed, nutrient loadings, time-series analysis

1 | INTRODUCTION

Karst landscapes drain approximately 12% of the earth's surface and serve as a drinking water source for nearly 25% of the Earth's population (Ford & Williams, 2007; Hartmann, Goldscheider, Wagener, Lange, & Weiler, 2014). Agricultural karst terrain is highly vulnerable to nutrient leaching, making karst landscapes hotspots for downstream eutrophication (Husic, Fox, Adams, et al., 2019; Jarvie et al., 2014; Mellander et al., 2013). There is increasing recognition that spring and surface water interfaces in karst landscapes are an ecosystem control point for biogeochemical transformations, yet these interfaces remain understudied (Briggs & Hare, 2018). In part, this is likely due to a lack of extensive databases for the management and prediction of karst water resources, as recently highlighted by Hartmann et al. (2014). Compounding these problems in temperate agroecosystems are complex seasonal patterns of nutrient dynamics both in the uplands (Exner-Kittridge et al., 2016; Ford, Williams, & King, 2018; Hartmann et al., 2014; Jarvie et al., 2014) and in-stream (Ford, Fox, Pollock, Rowe, & Chakraborty, 2015; Ford, Williams, & King, 2018; Shore et al., 2017) that require long-term data to capture variability in environmental drivers (Mulholland & Hill, 1997; Pionke, Gburek, Schnabel, Sharpley, & Elwinger, 1999; Stow, Cha, Johnson, Confesor, & Richards, 2015). The above information suggests that long-term datasets that can distinguish upland input dynamics from in-stream nutrient fate are needed to improve nutrient management in karst agroecosystems.

Regarding upland dynamics, recent advancements in techniques to quantify karst hydrologic pathways has led to a conceptualization of karst drainage as a series of reservoirs characterized by quick, intermediate, and slow flow pathways (Husic, Fox, Adams, et al., 2019; Husic, Fox, Ford, et al., 2019; Rimmer & Hartmann, 2014). Quickflow in karst watersheds reflects short residence time water originating from sinkholes, swallets, estavelles, and conduits (Pronk, Goldscheider, Zopfi, & Zwahlen, 2009; White, 2002). Intermediate flow pathways reflect moderate residence time and include recharge via soil macropores and epikarst fractures (Tzoraki & Nikolaidis, 2007; Williams, 2008). Slow flow pathways reflect longer residence times and convey water via diffuse Darcian recharge and percolation through small, low-permeability fissures in the epikarst (Ghasemizadeh et al., 2012; Mellander et al., 2013). Quantification of these pathway dynamics is now commonly performed using hydrograph separation techniques on flow time series (e.g., Husic, Fox, Adams, et al., 2019; Mellander et al., 2013).

Improved hydrologic characterization of karst watersheds has led to improved characterization of nutrient transport pathways. Nitrate concentration and subsequent leaching from the soil and epikarst zone have been found to control spring concentrations and loadings, whereas quickflow following storm events has been found to dilute concentrations relative to groundwater aquifer sources (Baran, Lepiller, & Mouvet, 2008; Husic, Fox, Ford, et al., 2019 and references within). Conversely, dissolved reactive phosphorus (DRP) has been found to be more concentrated in quickflow than in intermediate and slow flow pathways, stemming from agricultural management

practices in which high soil P levels at the soil surface become connected to the stream network through macropores and overland flow during quickflow (Baker, Johnson, & Confesor, 2017; Ford, Williams, Young, King, & Fischer, 2018; Jarvie et al., 2014; King, Williams, & Fausey, 2014). Despite this, karst conduits have recently been recognized to be highly retentive of DRP, even during quickflow, because of transient sediment deposits within preferential pathways that adsorb or settle a portion of reactive P (Jarvie et al., 2014; Mellander et al., 2013). Regarding seasonality of soil drainage waters during low flow in agroecosystems, a recent review suggests that N and P have been found to have contrasting seasonality with nitrate often having maximum concentrations in winter and minimum concentrations in summer, whereas DRP often has been found to have maximum concentrations in summer and minimum concentrations in winter (Ford, Williams, & King, 2018).

Regarding in-stream fate, agroecosystem streambeds often consist of exposed bedrock or thin surficial sediment deposits starkly contrasting the deep alluvial beds found in systems with predominant hyporheic zones (Fogle, Taraba, & Dinger, 2003; Ford & Fox, 2014; Husic, Fox, Agouridis, et al., 2017; Mellander et al., 2013). Recent studies have highlighted the impact of the surficial fine-grained laminae, in both surface and phreatic conduit streambeds, on nutrient attenuation and transient storage (Ford & Fox, 2014, 2015; Ford, Fox, & Pollock, 2017; Husic, Fox, Ford, et al., 2017; Husic, Fox, Agouridis, et al., 2017). Less is known about the role of exposed bedrock streambeds in low-gradient regions. Limited research in this area may stem from findings in steep-gradient bedrock systems that have shown less metabolic activity as compared with alluvial systems (e.g., Argerich, Haggerty, Marti, Sabater, & Zarnetske, 2011; Gibson, O'Reilly, Conine, & Lipshutz, 2015). Nevertheless, researchers have recognized that in-stream aquatic vegetation dynamics can influence transient storage and transformation of nutrients in exposed bedrock streams and thus should be considered in management strategies (Ensign & Doyle, 2006; Gibson et al., 2015; Hensley & Cohen, 2012).

Prevailing aquatic vegetative controls on nutrients in agroecosystem streams include autochthonous production of benthic algae and aquatic macrophytes, which has broadly led to their inclusion in existing surface water quality models (Chapra, Pelletier, & Tao, 2008; Mebane, Simon, & Maret, 2014; Park & Clough, 2012;). Vegetative nutrient dynamics in agroecosystem streams are often tied to benthic vegetation as opposed to production in the water column (Ford et al., 2017; Ford & Fox, 2014; Griffiths et al., 2012). Benthic aquatic macrophytes are rooted in streambed sediments and are thus unlikely to exert a strong control on nutrients in scoured bedrock channels (Park & Clough, 2012). However, non-rate-limiting nutrient conditions, open canopies, shallow streams, and low light attenuation results in the proliferation of thick benthic algae mats, of temporally varied persistence, in the stream network (Ford & Fox, 2014; Griffiths et al., 2012; Hotchkiss & Hall, 2015). These findings would tend to suggest that the prevalence of assimilatory and dissimilatory processes may be mediated by the presence of algal biomass in scoured bedrock channels. Assessing these perceptions in karst agroecosystems is critical given that karst drainages provide thermal buffering, have low

dissolved oxygen at low-flows, and deliver high nutrient loads that stimulate in-stream biochemical reactivity (Dirnböck, Kobler, Kraus, Grote, & Kiese, 2016; Husic, 2018; Husic, Fox, Adams, et al., 2019; Husic, Fox, Ford, et al., 2019; Jiang, Hu, & Schirmer, 2013; Knierim, Pollock, Covington, Hays, & Brye, 2017; Mahler & Bourgeois, 2013).

This study aims to quantify hydrologic and in-stream aquatic vegetation controls on nutrient dynamics in a karst agroecosystem watershed through analysis and statistical modelling of long-term hydrologic and water quality data. Specific objectives of this study were to (a) assess spatiotemporal dynamics of nutrients in a karst agroecosystem using long-term measurements from nested spring and stream sites, (b) quantify flow and nutrient pathway dynamics in a karst watershed and assess their influence on nutrient loading, and

(c) assess the fate of nutrients in bedrock-controlled surface streams draining karst agroecosystems.

2 | METHODS

2.1 | Study site

To meet the objectives of this study, the Camden Creek watershed (Figure 1) was selected as the study site. Camden Creek drains a large portion of the University of Kentucky Animal Research Center (ARC) farm and lies within the Inner-Bluegrass physiographic region of central Kentucky. The region is characterized by a temperate Midwestern

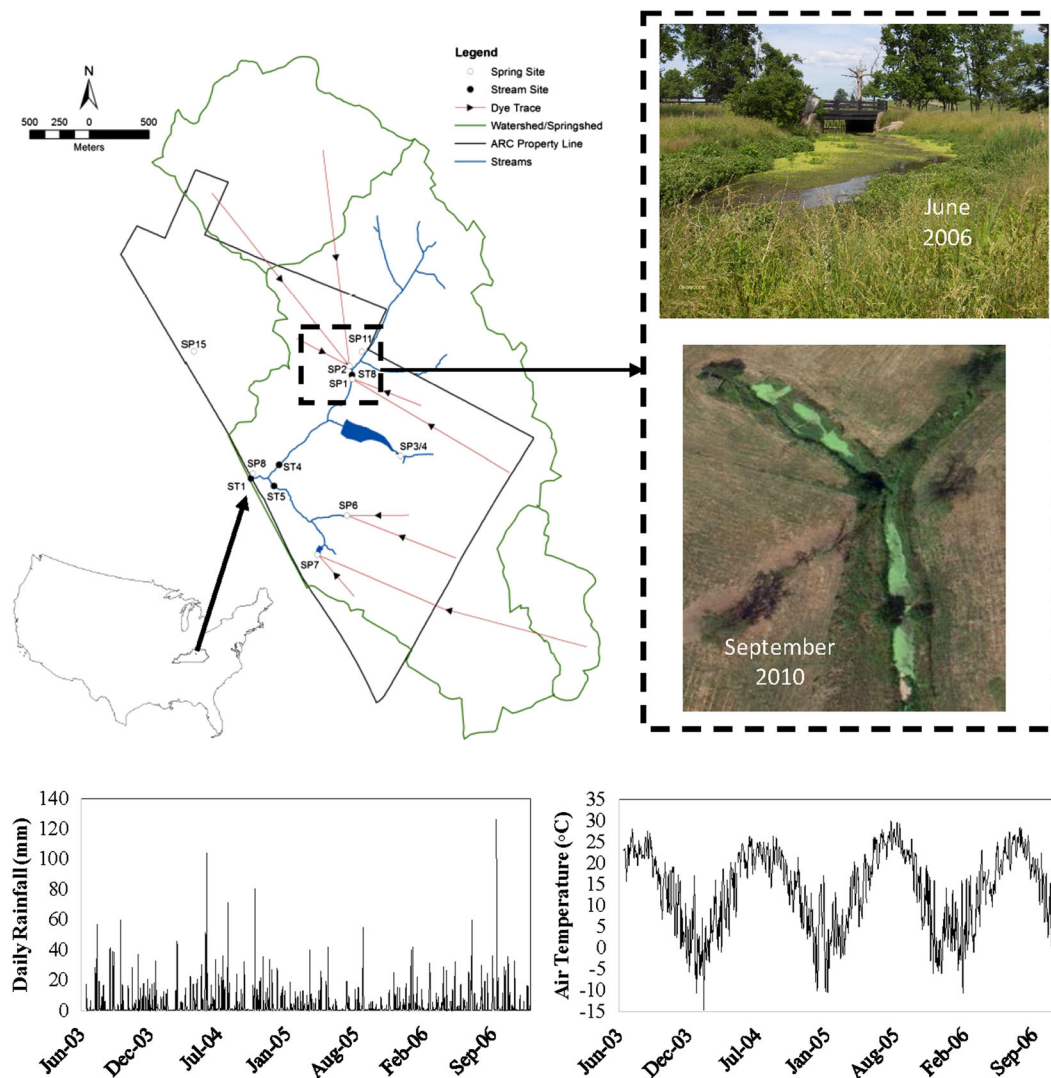


FIGURE 1 Watershed delineation, site locations, imagery, and weather data from the Camden creek watershed in central KY, USA. The inner polygon represents the surface watershed delineation for Camden Creek, and the outer polygons reflect additional spring shed drainage areas to sinkholes connected to the Camden Creek Watershed. Spring (SP1, SP2, SP3/4, SP6, SP7, SP8, and SP11, and SP15) and in-stream (ST1, ST4, ST5, and ST8) sampling locations are identified. Site images show prevalence of floating aquatic macrophytes (primarily duckweed) to persist during the growing season. The top image is a picture from June 2006, and the bottom image was obtained from Google Earth and was taken September 2010. The bottom charts show temperature and rainfall data collected at a NOAA gauging station located within the ARC farm from June 2003 to June 2006 (overlapping with nutrient data collected in this study)

U.S. climate with four distinct seasons including moderately cold winters, warm and humid summers, and moderate transition periods in spring and fall (see Figure 1). Average annual rainfall monitored at the ARC from June 2003 to December 2006 was 1,253 mm year⁻¹, and average annual temperature was 13.3°C.

The watershed is characterized by broad, shallow sinkholes; low-relief, broad valleys and ridges; sparse rock outcrops, and thick, fertile, phosphatic Ordovician limestone; and shale-residual soils, typical of the region. The USDA soil classification of the ARC shows predominantly Hagerstown and Maury silt loam soils. Soil depths vary from less than 1 m on valley walls to 5 m on ridge tops. The hydrogeology of the study site has been previously characterized (Keagy, Dinger, Fogle, & Sendlein, 1993), and we briefly summarize key components. The study area is almost entirely underlain by Ordovician-age rocks, predominantly Lexington limestone with nearly horizontal strata. Limestone members include (in descending order) Devils Hollow, Tanglewood, Brannon, and Grier, with most soils having formed over the Tanglewood member. The regional water table exists within the Grier member, and perched aquifers have been observed from well nests both above the bedrock surface and within the Grier member.

Fourteen springs have been identified on the ARC, including ephemeral ($n = 9$), local perennial ($n = 3$), and regional perennial ($n = 2$) springs (Keagy et al., 1993). Dye traces were conducted in the watershed on regional perennial springs and highlight sinkhole connectivity to create a cumulative watershed drainage area of 1,069 ha, which extends well beyond the surface watershed drainage area of 771 ha (Keagy et al., 1993; Reed, McFarland, Fryar, Fogle, & Taraba, 2010; Figure 1). Heterogeneity in karst conduit maturity has been previously documented in the region (e.g., Mahoney, Fox, & Al Aamery, 2018; Reed et al., 2010). We classify the Camden Creek watershed to have immature karst as compared with other basins in the region. For example, Reed et al. (2010) highlighted that a regional spring less than 5 km away, draining a similar area to the Camden Creek watershed, had more complex architecture and a resurgent stream channel, contrasting regional springs in Camden Creek. Further, the abundance of local ephemeral springs that drain in upper portions of the epikarst, as opposed to connecting to deeper conduits at the regional level, suggests less developed karst conduit architecture, broadly, in the watershed.

During the timeframe of this study, portions of the ARC were used for precision and site-specific agriculture operations, as well as tobacco, row crops, small grains, and animal research plots. Both organic and inorganic fertilizers were applied to the crop production systems throughout the monitoring period. Tobacco received both N and P inorganic fertilizer sources whereas the other row crops (e.g., maize) were side-dressed with inorganic N. Surrounding landscapes were predominantly horse pasture with some sparse residential development. The streams at the ARC are shallow, flow over limestone bedrock with limited sediment deposits, and are generally unshaded with some riparian vegetation (Fogle et al., 2003). Sediment storage in the streambed is low with most of the bedrock exposed. Low storage of sediment in the streambed likely reflects flow pathways because most stormflow enters the streams via springs as opposed to overland flow

(Keagy et al., 1993), thus limiting erosion and transport of sediment to the stream. For this reason, we did not explicitly consider the role of fluvial sediment deposits to influence nutrient processes in this study.

2.2 | Data collection and laboratory analysis

Data collection was conducted at both stream (ST) and spring (Sp) locations in the watershed. Monitored stream sites included ST1 (watershed outlet), ST5 (major tributary draining into the main stem), ST4 (main-stem site that was upstream of the confluence of the main stem with ST5), and ST8 (upstream boundary of the main stem). Several spring sites were monitored both within the Camden Creek watershed (Sp1, Sp2, Sp3/4, Sp6, Sp7, Sp8, and Sp11) and outside of the watershed boundary but within the ARC farm (Sp15; Figure 1).

Grab sampling began in October 1996 and was conducted through June 2007. Unpreserved samples were collected for NO₃⁻ in 250-ml amber glass bottles. A secondary unpreserved DRP split was collected in 125-ml polyethylene bottles. Preserved (H₂SO₄) samples were collected in 250-ml clear glass bottles for total ammoniacal-N (TAN) and total organic C (TOC). All samples were placed on ice immediately following sampling and were delivered to the labs within 6 hr of collection. The DRP split was filtered prior to delivery to the lab. Samples were then refrigerated prior to analysis and were analysed within standard preservation windows. Regarding laboratory analysis, NO₃⁻, TAN, and TOC were analysed in the Kentucky Geological Survey laboratory. A 50-ml split from the 250-ml amber glass bottle was filtered and analysed on a Dionex Ion Chromatograph for NO₃⁻ within 48 hr of sample collection. TOC was analysed by a UV peroxide instrument by Phoenix on the unfiltered H₂SO₄ preserved sample within 28 days. TAN was determined by filtering a 50-ml split of the H₂SO₄ preserved sample, adjusting the pH between 9 and 11 and analysing for NH₃-N colorimetrically using a UV Vis spectrometer by Varian within 28 days of sample collection. Orthophosphate (DRP) was determined colorimetrically at 630 nm for a 1-ml filtered split within 28 days of sample collection (D'Angelo, Crutchfield, & Vandivere, 2001).

The silt-loam soil texture and abundance of sinkholes at the study site promoted streamflow to be predominantly initiated by spring discharge from the 14 spring sites at both low and high flow conditions, hence annual nutrient fluxes from springs were a large fraction of watershed fluxes. We aimed to compare upland spring contributions of nutrients with fluxes at stream monitoring stations to identify the influence of in-stream processes. Spring sites were generally sampled once every 4 weeks, and stream sites were sampled biweekly. TOC and TAN were collected at all stream sites throughout the monitoring period but only at Sp2 and Sp6 from 1997 to 1999. Long-term records from the spring sites were primarily from the local perennial and local ephemeral systems (Sp1, Sp3/4, Sp6, Sp8, Sp11, and Sp15) reflecting the wide range of variability in spring nutrient inputs on the farm. The regional perennial springs (Sp2 and Sp7) were sparsely monitored directly due to logistical issues of monitoring at the spring orifice. The ephemeral nature of many of the monitored springs resulted in periodic discontinuities in the individual spring datasets, limiting the ability

to perform time-series analysis on each. Local ephemeral springs were typically activated during stormflow and recession of perched aquifers in the vadose zone. Local perennial springs breached confining layers via sinkholes, allowing perched aquifers to drain at a more regional level and hence flowed year-round (Keagy et al., 1993). Flowrate from each contributing spring was not available to perform flow weighted averaging. Nevertheless, upon initial investigation, we found that visual time series of maximum, minimum, and averaged nutrient concentrations for spring sites displayed analogous temporal dynamics (e.g., seasonal variability). In order to generate a continuous dataset of composite spring contributions from the watershed and subsequently make downstream comparisons, we averaged values from spring sites on days that spring data was available (Sp_{av}). Because a large proportion of the springs were monitored, it is reasonable to use the central limit theorem to approximate the composite spring nutrient concentrations for the watershed by averaging the individual spring contributions at a given point in time (Olkin, Gleser, & Derman, 1994).

Flowrate was monitored at ST1 from October 1996 through September 2006 using an ISCO 4220 Flow Meter with a submerged pressure transducer. From October 1996 through August 2003, depth (d) upstream of a broad-crested weir was used to estimate flowrate (Equation 1) per the Teledyne flow measurement handbook equation for a broad crested weir.

$$Q = -4.769d + 49.299d^{1.5}, \quad (1)$$

where Q was volumetric flowrate ($\text{ft}^3 \text{ s}^{-1}$), and d was the depth (ft). In September 2003, a 120° V-notch weir was installed at ST1. Flow was estimated using a piecewise function considering flow below (Equation 2) and above (Equation 3) the top of the weir plate.

$$Q = 4.330d^{2.5} \text{ if } d < 1 \text{ ft}, \quad (2)$$

$$Q = 4.330 + 46.62(d-1)^{1.5} \text{ if } d > 1 \text{ ft}. \quad (3)$$

2.3 | Exploratory statistical analysis of 10-year nutrient dataset

Nonparametric statistical tests and exploratory visual statistical plots were utilized to compare nutrient datasets between stream monitoring stations and the averaged spring dataset. Box-and-whisker plots were used to visualize distributions and were generated in *SigmaPlot* 13 (Systat Software, Inc). Given the skew of the data, y-axes were plotted on a log scale, and nonparametric statistical tests were performed using *SigmaStat* 13. Specifically, the nonparametric Kruskal–Wallis one-way analysis of variance test was utilized to determine statistically significant ($\alpha = 0.05$) differences in median values between groups. For parameters with significant differences between groups, a post hoc test using Dunn's method was performed to compare significant differences between data pairs.

Time-series analysis was used to elucidate information on statistically significant trends in the nutrient species. Given that the dataset was collected biweekly to monthly, over a decade, we expected statistically significant fluctuations to reflect prevailing seasonal fluctuations of spring and in-stream dynamics. Empirical mode decomposition (EMD) was selected as the preferred method for the analysis because the method is purely empirical (i.e., does not use prescribed oscillating functions), can be applied to a wide class of signals, and can be used as an a posteriori approach, which is ideal for an exploratory analysis (Ford et al., 2015; Ford, Williams, & King, 2018; Huang et al., 1998; Wu, Huang, & Peng, 2007). EMD was conducted utilizing a six-step iterative procedure as summarized in Ford et al. (2015) in which a dataset is decomposed into a series of intrinsic mode functions (IMFs) and a residual term. IMFs are a finite series of amplitude- and frequency-modulated, oscillatory functions. The lowest frequency IMF is identified as the base trend, and the highest frequency IMF is considered noise for well-sampled datasets (Wu et al., 2007). We compiled a previously published code in Matlab R2016a that performs EMD and generates IMFs for each site (Rato, Ortigueira, & Batista, 2008). We performed statistical significance evaluations to test the hypothesis that IMFs of the dataset were different from white noise IMFs. Statistically significant trends were determined visually by variance–frequency plots in which IMFs with variance greater than three times the standard deviation for noise at a particular frequency were identified as significant trends (see Ford et al., 2015). A further explanation of the method is detailed elsewhere (e.g., Ford et al., 2015; Wu et al., 2007; Wu & Huang, 2004). Statistically significant IMFs with frequencies between 0.7 and 1.3 years were considered seasonal IMFs because fluctuations may not have been pronounced in some years (frequency > 1 year) or may have multiple maxima and minima in a year (frequency < 1 year).

2.4 | Multiple linear regression analysis

Given the importance of downstream NO_3^- and DRP transport, we performed multiple linear regression (MLR) modelling using presumed important drivers of nutrient dynamics to create a continuous predictive model of load estimates. MLR was performed on the average spring (Sp_{av}) and watershed outlet (ST1) datasets. In terms of independent variables in the MLR, seasonal fluctuations in environmental parameters (e.g., atmospheric variables and soil properties) and precipitation-driven hydrologic variability have been identified as key factors controlling nutrient concentrations in agroecosystem watersheds (Ford, Williams, & King, 2018 and references within; Sinha, Michalak, & Balaji, 2017). Seasonal maxima and minima often show hysteresis from common predictors such as temperature, light availability, pH, and dissolved oxygen (e.g., Ford et al., 2015; Ford & Fox, 2014; Ford, Williams, & King, 2018). Upon initial investigation, we found a power relationship between day of year and NO_3^- concentration at spring sites, which reflects previously reported timing of seasonal source nutrient concentration dynamics in temperate agroecosystems (Ford, Williams, & King, 2018). Day of year was used

as a response variable to account for the presence or absence of the upland seasonal trend. Regarding hydrologic impacts, we anticipated both log-linear relationships (Schilling & Lutz, 2004) and linear relationships (e.g., Zheng et al., 2015) between flowrate and nutrient concentrations. Given the multiple descriptors, we performed an MLR with the following structure for all nutrient species:

$$C_{pre}^{i,j} = \beta_0^{i,j} + \beta_1^{i,j}Q + \beta_2^{i,j}\ln(Q) + \beta_3^{i,j}Day + \beta_4^{i,j}(Day^2), \quad (4)$$

where, β 's were the regression coefficients, i was the nutrient species, j was the site identifier (either Sp or $ST1$), C_{pre} was the predicted nutrient concentration, Q was the flowrate at the watershed outlet, and Day was the day of year (ranging from 1 to 365). The regression analysis was performed in Matlab R2016a using the built-in MLR analysis function (*fitlm*). The procedure identified coefficients that provided a "best fit" linear model and provided a suite of statistical data. Specifically, we used the F statistic to test the null hypothesis that individual coefficients were not equal to zero and the null hypothesis that the overall MLR model provided a superior fit to a mean trend. The p values were calculated for the F statistics in both hypothesis testing scenarios, and significance results for $p < 0.10$, $p < 0.05$, $p < 0.01$, and $p < 0.001$ were provided. Acceptance of the null hypothesis occurred for $p < 0.10$. If individual parameters were found insignificant, they were removed from the MLR, and the regression was rerun until all parameters were significant and the overall model was significant.

2.5 | Pathway loading analysis

We conducted a loading analysis to quantify the relative importance of flow pathways and in-stream processes for watershed nutrient loading. Loads were estimated for nitrate and DRP at $ST1$ and Sp_{av} using continuous flow data and regression-predicted concentration values as follows:

$$Load^{i,j} = \sum_{t=1}^n Q_t^{ST1} C_{pre-t}^{i,j} \Delta t, \quad (5)$$

where $Load$ is the mass loading (kg), t is the time index, n is the number of timesteps in the analysis, and Δt is the timestep. Although our focus for the pathway analysis was on loadings at $ST1$, we used loadings estimated with Sp_{av} regression results for the in-stream fate analysis described in Section 2.6.

To estimate the contribution of flow pathways to nutrient loading dynamics, we multiplied the load from the regression model at $ST1$ by the fraction of flow from karst reservoir pathways, which were estimated continuously using hydrograph separation. In lieu of event-based high-resolution data, we were unable to perform a loadograph recession (e.g., Husic, Fox, Adams, et al., 2019; Mellander et al., 2013); however, given the longevity (10 years) of data collected during a wide array of hydrologic conditions, we were able to perform hydrograph separation. Specifically, we developed a master recession curve to identify and assess prominent pathway reservoirs. Thereafter, we performed hydrograph separation of all individual events to create

a continuous time series of discharge, providing a tool to assess temporal loading dynamics from each reservoir.

Forty-one recessions with a duration of at least 15 days were identified during the 10-year study period. Of the 41 recessions, 33 were used to generate the master recession curve. The eight discarded recessions were either (a) primarily comprised of days with zero flow (i.e., no flux) or (b) had nonlinear reservoir recessions likely associated with later rainfall disrupting the initial recession. Thereafter, the 33 individual recessions were organized such that they aligned to create a single, master recession using the RC 4.0 software (HydroOffice; Malík & Vojtková, 2012). The y-axis was changed to a logarithmic scale, and distinct linear segments of the master recession curve were identified. Recession coefficients (α) were then manually calibrated to generate the best-fit solution of discharge reservoirs to the master recession curve. Lastly, the areas under the quickflow and slow flow curves were integrated to calculate the approximate contribution of each pathway to net water drainage.

A master recession curve is useful for identifying drainage reservoirs, quantifying long-term contributions from dynamic transfer zones, and estimating recession coefficients for modelling purposes. However, the master recession curve does not provide a continuous time-series estimate of flow contribution from each reservoir. To address this problem, individual hydrograph separation was performed for each storm event during the 10-year study period. Hydrograph separation was performed with the following steps: (a) a "storm event" was first defined as a period of hydrologic activity resulting in a rapid increase in discharge, followed by a gradual recession that ends when the next spike in flow was observed; (b) a linear increase in slow flow was assumed from the beginning of the rising limb of the hydrograph to the inflection point on the falling limb, and this point signifies the end of quickflow (Husic et al., 2019a); and (c) event contribution by each pathway was calculated as the area between two curves.

2.6 | In-stream vegetation dynamics

We aimed to assess the perception that autotrophic algal biomass dynamics controlled in-stream nutrient fate in the bedrock-controlled streambed. Recent modelling and isotope monitoring approaches have shown newly fixed autochthonous C may have short residence time due to respiration or organic matter exudation resulting in regeneration of available nutrients to the stream channel or have longer-term storage in algal biomass that is subsequently subjected to downstream export due to hydraulic scour and sloughing (Ford & Fox, 2014; Hotchkiss & Hall, 2015). Although we did not have measurements or modelling of algae mass transfer dynamics from the Camden Creek watershed, previously published modelling results of algae C dynamics in the nearby South Elkhorn watershed were used to estimate fluxes on a monthly basis (Ford & Fox, 2014). Briefly, the previously published algae growth and decomposition model (Ford & Fox, 2014; Rutherford, Scarsbrook, & Broekhuizen, 2000) estimates C fixation ($gC\ m^{-2}\ day^{-1}$) as a function of light, temperature, and population saturation limitations, estimates biochemical losses ($gC\ m^{-2}\ day^{-1}$) using a

calibrated first-order equation that varies as a function of temperature (Rutherford et al., 2000), and estimates downstream export ($\text{gC m}^{-2} \text{ day}^{-1}$) as a result of physical scouring of algal biomass. The biochemical loss term is assumed to encompass losses associated with direct heterotrophic respiration of detrital material, endogenous respiration, and leaching of algal exudates; hence, the fate and partitioning of this lumped pool is uncertain and is addressed using a scenario analysis described below. Although the model also explicitly accounts for heterotrophic organic matter breakdown and processing of the particulate detrital algal pool, these fluxes were found to be insignificant for dissolved nutrient considerations and hence were not included to simplify the analysis. Detailed information regarding inputs, parameterization, and model evaluation are provided in Ford and Fox (2014).

We utilized reach-averaged results from a 5-year (January 2006 through December 2010) model simulation in the South Elkhorn watershed. Specifically, we used estimates of C fixation by algae, biochemical losses from the algal pool to the stream water, and downstream export. Results from the model were composited on a monthly basis to account for annual variability in processes (e.g., wet vs. dry summers), which enabled direct comparison with monthly averaged differences between the spring and watershed outlet load estimates. Because the model predictions estimate C fluxes, we used a typical C:N:P ratio for algal biomass (40:7.2:1) and assumed biochemical processes had analogous stoichiometry, which is assumed in other algal nutrient models (Chapra et al., 2014).

To assess if benthic algae fluxes could help inform differences between loads from ST1 and Sp_{av} , we compared the ratio of monthly flow-weighted concentrations for the baseline regression model with several hypothetical scenarios. The ratio was quantified as follows:

$$\frac{CSp_{av}}{CST1} = \frac{CSp_{av} \cdot Q}{CST1 \cdot Q} = \frac{\text{Load } Sp_{av}}{\text{Load } ST1} \quad (6)$$

If the ratio was equal to one, this signified that spring concentrations (and loading) were equal to that of the watershed outlet (i.e., conservative transport with no additional inputs). If the ratio was greater than one, the spring contributions were either diluted by another flow source (e.g., overland flow) or nutrients were attenuated within the stream channel resulting in higher nutrient concentrations at the spring than at the watershed outlet. If the ratio was less than one, the spring contributions were either mixed with a concentrated source or net nutrient regeneration occurred within the stream channel resulting in lower concentrations at the spring relative to the watershed outlet. Because the Sp_{av} concentrations were arithmetic averages of all springs on the site, we did not expect values to equal 1; however, we did anticipate relatively stable ratios throughout the year if nutrients behaved conservatively in the channel.

Because the algae growth and decomposition model did not explicitly simulate organic and inorganic N and P fate following leaching or respiration from the algal pool, we ran three hypothetical scenarios for fate of the biochemical loss pools and adjusted the spring loadings accordingly to investigate potential changes in Equation 6. Scenario 1 assumed all algae biochemical losses were associated with excretion of

organic matter exudates as dissolved organic matter (i.e., dissolved organic carbon, dissolved organic nitrogen, and dissolved organic phosphorus were regenerated to the stream channel). Scenario 2 assumed that all algae biochemical losses were a result of respiration or that organic excretions were rapidly mineralized to ammonium and DRP without nitrification occurring (i.e., TAN and DRP were regenerated to the stream channel). Scenario 3 was the same as Scenario 2, except we assumed that all TAN was subsequently nitrified within the stream channel (i.e., NO_3^- and DRP were regenerated to the stream channel). Although these scenarios represent extreme conditions, they provided a spectrum of potential fates for biochemical losses for N and P.

3 | RESULTS

3.1 | Exploratory analysis of the 10-year dataset

The importance of in-stream production of TOC and TAN and removal of NO_3^- and DRP are recognized by the visual box and whisker plots and statistically significant differences from the spring inputs to downstream gradient stream sites (Table 1 and Figure 2). Generally, we found statistically significant increases of TOC and TAN and decreases of DRP and NO_3^- from upstream springs (Sp_{av}) to downstream (ST1, ST4, ST5, and ST8) locations. As well, we found that increasing drainage areas for stream sites resulted in increasing TOC and TAN and decreasing DRP and NO_3^- . However, some exceptions were observed, for instance when comparing ST4 and ST8 of NO_3^- or comparing ST1 with ST8 for DRP in which significant median differences were not observed between sites. These deviations are likely associated with source mixing of two tributaries or a lack of in-stream influence during median flow conditions.

Statistically significant IMFs from the EMD were primarily governed by seasonal trends and provided insight into timing of seasonal maximum and minimum nutrient concentrations (Table 2; Figure 3). For TOC, NO_3^- , and TAN, all stream datasets had statistically significant seasonal IMFs. For DRP, only ST4 and ST5 had significant seasonal fluctuations. Regarding seasonality of Sp_{av} for TAN and TOC, insufficient length of data was available to perform EMD. However, data were available throughout at least one full calendar year, and variability for both parameters were low relative to stream sites; hence, seasonal variability was not expected to be prominent (Figure 2). Spring DRP and NO_3^- concentrations had sufficient data for EMD analysis, and we found significant seasonality for NO_3^- but not for DRP (Table 2). Regarding timing of maxima and minima, TOC and TAN generally were maximum in summer (July–September) and minimum in winter (January–March). Similar to findings in Figure 2, NO_3^- and DRP IMFs (Figure 3) showed inverse relationships to TOC and TAN in which maximums generally occurred in winter (January–March) and minimums generally occurred in summer (July–September). Of note was the seasonal NO_3^- IMF for the Sp_{av} data, which had analogous timing to stream sites for maximum concentrations (January–March) but differed for minimum concentrations and

TABLE 1 Statistical significance test results comparing nonparametric distributions of stream sites and average spring values was performed using the Kruskal–Wallis one-way analysis of variance with a post hoc multiple comparison procedure using Dunn's test

Kruskal–Wallis one-way analysis of variance on ranks					
Nutrient	TOC	DRP	NO ₃ [−]	TAN	
<i>p</i> value	<0.001	<0.001	<0.001	<0.001	
Post hoc Dunn's Test for difference between group pairings					
TOC	ST1 (n = 258)	ST4 (n = 201)	ST5 (n = 202)	ST8 (n = 247)	Sp _{av} (n = 68)
ST1 (n = 258)	1				
ST4 (n = 201)	>0.05	1			
ST5 (n = 202)	>0.05	<0.001	1		
ST8 (n = 247)	<0.001	<0.001	>0.05	1	
Sp _{av} (n = 68)	<0.001	<0.001	<0.001	<0.001	1
DRP	ST1 (n = 264)	ST4 (n = 202)	ST5 (n = 199)	ST8 (n = 254)	Sp-Av (n = 166)
ST1 (n = 264)	1				
ST4 (n = 202)	0.032	1			
ST5 (n = 199)	0.007	<0.001	1		
ST8 (n = 254)	>0.05	<0.001	<0.001	1	
Sp _{av} (n = 166)	<0.001	<0.001	<0.001	<0.001	1
NO ₃ [−]	ST1 (n = 267)	ST4 (n = 191)	ST5 (n = 206)	ST8 (n = 248)	Sp-Av (n = 165)
ST1 (n = 267)	1				
ST4 (n = 191)	<0.001	1			
ST5 (n = 206)	<0.001	<0.001	1		
ST8 (n = 248)	0.002	>0.05	<0.001	1	
Sp _{av} (n = 165)	<0.001	<0.001	<0.001	<0.001	1
TAN	ST1 (n = 163)	ST4 (n = 90)	ST5 (n = 92)	ST8 (n = 98)	Sp-Av (n = 36)
ST1 (n = 163)	1				
ST4 (n = 90)	N/A	1			
ST5 (n = 92)	N/A	>0.05	1		
ST8 (n = 98)	>0.05	0.011	N/A	1	
Sp _{av} (n = 36)	0.046	0.007	>0.05	N/A	1

Note. Values reported are the *p* values from the Kruskal–Wallis and Dunn's tests. A significance level of 5% ($\alpha = 0.05$) was chosen to test for significance. Groups that are statistically differentiable are identified in **bold** and *italics*. Statistical tests were performed in SigmaPlot 13.0. Some comparisons were unable to be performed and are denoted using N/A. DRP: dissolved reactive phosphorus; TAN: total ammoniacal-N; TOC: total organic C.

occurred in June–July. In some years, ST1 for NO₃[−] was observed to be out of phase from the other spring and stream sites, likely reflecting the influence of mixing dynamics between ST4 and ST5 at a confluence immediately upstream of the watershed outlet (Figure 1).

3.2 | MLR analysis

Results of the MLR analysis for spring and stream sites provided significant overall predictive models for NO₃[−] and DRP concentrations but showed variable significance regarding presumed important drivers (Table 3). The regression model for DRP described 44% of the variability in the concentration data at the watershed outlet and

11% at Sp_{av}. DRP had significant coefficients for the linear flow term and the overall regression model ($p < 0.001$) for Sp_{av} and significant values for the linear and logarithmic flow coefficients and the overall regression model ($p < 0.001$) at ST1. Regarding NO₃[−], the regression model described 63% of the variability in the concentration data at the watershed outlet and 4% at Sp_{av}. For Sp_{av}, NO₃[−] had significant coefficients for the linear flow term, the seasonal power relationship, and the overall regression model ($p < 0.1$). For ST1, NO₃[−] had significant values for all coefficients and the overall regression model ($p < 0.001$).

Visual comparisons of the measured data and regression model outputs at ST1 for the major macronutrients (i.e., NO₃[−] and DRP) highlight the importance of flowrate and, to a lesser degree, seasonality of

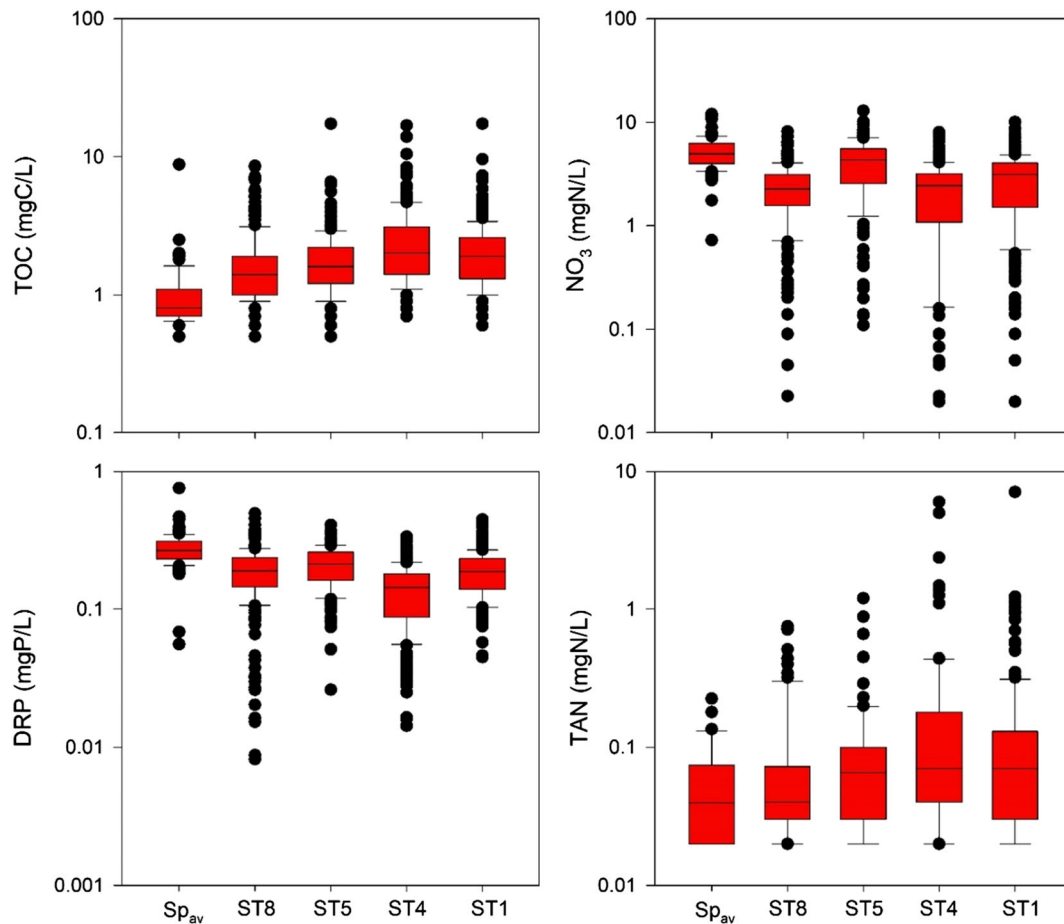


FIGURE 2 Box and whisker plots of nutrient concentration data from stream sites (ST1, ST4, ST5, and ST8) and the average spring values (Sp_{av}). Boxes show the inner-quartile range (25th and 75th percentiles) and median (50th percentile) values, whereas whiskers show 5th and 95th percentile values. Values greater or less than 5th and 95th percentiles are included as points

vadose zone nutrient concentrations to influence watershed nutrient concentrations in the study watershed (Figure 4). Generally, we found that the positive log-linear relationship between flowrate and DRP controlled at low flows ($<0.7 \text{ m}^3 \text{ s}^{-1}$), and the positive linear relationship became the governing factor for greater flowrates (Table 3; Figure 4a). We found a positive log-linear relationship between flowrate and NO_3^- concentration at the watershed outlet when flowrate was less than approximately $0.6 \text{ m}^3 \text{ s}^{-1}$ and prominence of a negative linear relationship between flowrate and NO_3^- at flowrates greater than that flow condition (Table 3; Figure 4c). We recognized the importance of the upland seasonal term for ST1 given the noise in the regression model when plotting flowrate against NO_3^- , which was not observed for DRP (Figure 4a,c). The timing of the seasonal minimum (based on coefficients in Table 3 and the superimposed significant seasonal trend in Figure 4d) aligns well with minimum values from the seasonal IMF for Sp_{av} at the beginning of summer, highlighting the importance of seasonality of concentrations in the vadose zone to affect NO_3^- concentration and loading at the watershed outlet. However, the overwhelming seasonal trend for both DRP and NO_3^- appear to be associated with flow controls in which low flow conditions correlate with DRP and NO_3^- concentrations and are

consistent with the timing of minimum values in the EMD analysis for the stream sites.

3.3 | Pathway loading analysis

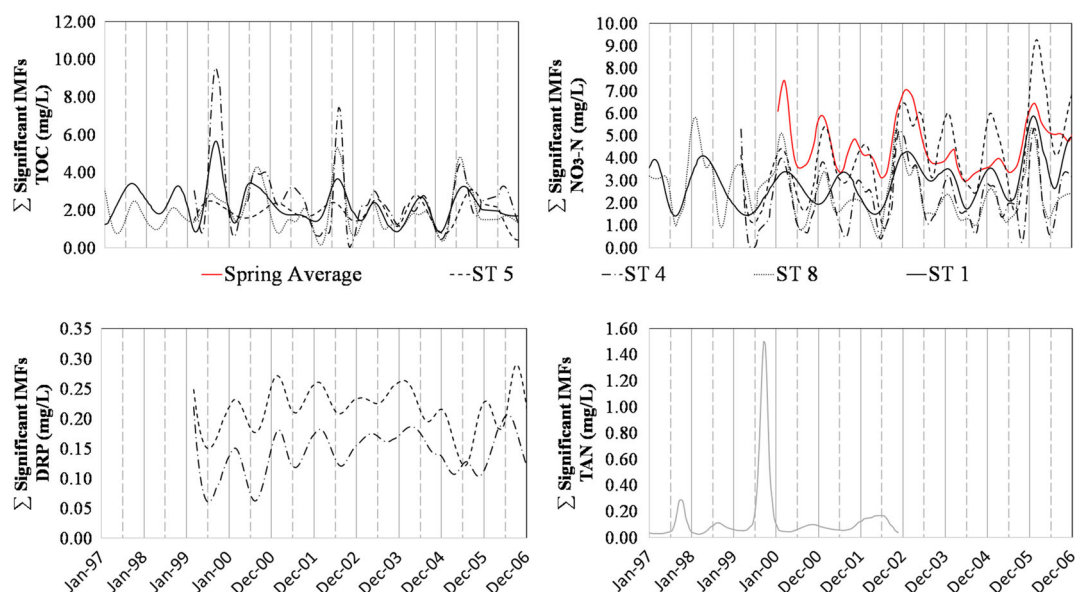
Results from the master recession curve at the watershed outlet identifies two distinguishable slopes, supporting a two-reservoir (R1 for steep and R2 for mild) karst drainage network. The recession coefficients (α) for R1 and R2 were 0.25 and 0.07 day^{-1} , respectively (Figure 5). On average, 25% of the discharge is drained by R1 whereas the remaining 75% is drained by R2 (Table 4). Results from the continuous recession analysis of the 10-year dataset show average flowrates of 0.043 and $0.133 \text{ m}^3 \text{ s}^{-1}$ for R1 and R2, respectively (Table 4). Seasonally, flowrate from both reservoirs was greatest in the winter and least in the summer (Table 4, Figure 6). Fractional contributions of reservoirs deviated from maximum and minimum timing of flowrates, with R1 being highest in summer and lowest in spring, and R2 being highest in spring and lowest in summer (Table 4), highlighting the flashiness of the system during dry summer months.

Results from the nutrient pathway analysis provide quantitative estimates of R1 and R2 loadings from the watershed (Table 4). Nitrate

TABLE 2 Frequency (years) of statistically significant intrinsic mode functions (IMFs) from the empirical mode decomposition analysis of monitoring sites

Parameter	Sp_{av}	ST1	ST4	ST5	ST8
Total organic carbon	N/A	1.19	0.61	1.38	0.83
		1.79	1.10		1.26
		3.58	1.65		1.95
		5.38	5.5		3.58
					5.38
Dissolved reactive phosphorus	No significant IMFs	No significant IMFs	0.97	0.97	No significant IMFs
			2.75	2.75	
			4.13	4.13	
Nitrate	0.68 1.18 2.17 4.33	1.26	0.59	0.93	0.72
		2.69	1.34	1.85	1.19
			1.83	3.33	1.65
			2.75		2.69
			5.5		
Total ammoniacal nitrogen	N/A	1.20	N/A	N/A	N/A

Note. Statistically significant trends were determined visually by variance–frequency plots in which IMFs with variance greater than three times the standard deviation for noise at a particular frequency were identified as significant trends (see Ford et al., 2015). Residual long-term trends are not included in this table because the timescale of their fluctuations are unknown but are reflected in the sum of statistically significant fluctuations in Figure 3. N/A is used for sites where the specified constituent was not measured.

**FIGURE 3** Results of the empirical mode decomposition (EMD) time-series analysis showing statistically significant intrinsic mode functions (IMFs) for Sp_{av} , ST1, ST4, ST5, and ST8 sites. Results are provided for a 10-year span from January 1, 1997 to December 31, 2006. Not all data sets were obtained for a long enough record for time-series analysis (e.g., total organic carbon [TOC] and total ammoniacal-N [TAN] at spring sites were collected for 2 years). Solid lines represent the beginning of the year (January 1), and dashed vertical lines represent the middle of the year (July 1). Note the EMD analysis of Sp_{av} are only shown for 2000–2006 because data was collected at a biweekly temporal resolution for some springs prior to 2000

loadings from R1 and R2 generally followed the same seasonal trends as flowrate; however, the fractional contribution of NO_3^- from R1 was consistently less than the fractional contribution of R2, suggesting lower flow-weighted concentrations of NO_3^- from R1. On average, we estimate a load of $9.64 \text{ kgN day}^{-1}$ from R1 and $40.85 \text{ kgN day}^{-1}$ from R2. As a result, nearly 20% of nitrogen was associated with R1 and 80% with R2. Regarding DRP, we found a load of $1.11 \text{ kgP day}^{-1}$ from R1 and $2.69 \text{ kgP day}^{-1}$ from R2. As a result, nearly 30% of

phosphorus was associated with R1 and 70% with R2. Contrasting NO_3^- , this result showed that R1 had higher flow-weighted DRP concentrations as compared with R2.

3.4 | In-stream vegetation scenarios

Application of stoichiometric relationships to the algal C model from the neighbouring stream highlight the importance of algal dynamics

TABLE 3 Results of the multiple linear regression analysis. Values denote estimates with standard error in parenthesis

Spring	DRP	NO ₃ ⁻
β_0	0.21 (0.02)****	5.3 (0.60)****
β_1 (Q)	0.14 (0.03)****	1.7 (0.87) ^{a*}
β_2 (ln(Q))	-9.1×10^{-3} (5.5×10^{-3})	-0.22 (0.15)
β_3 (Day)	1.8×10^{-4} (2.3×10^{-4})	-0.01 (6.4×10^{-3})**
β_4 (Day ²)	-3.9×10^{-7} (5.9×10^{-7})	3.9×10^{-5} (1.6×10^{-5})**
Stream	DRP	NO ₃ ⁻
β_0	0.24 (0.01)****	7.3 (0.29)****
β_1 (Q)	0.07 (0.02)***	-2.26 (0.45)****
β_2 (ln(Q))	0.02 (3.4×10^{-3})****	0.77 (0.07)****
β_3 (Day)	-1.1×10^{-4} (1.4×10^{-4})	-0.02 (2.9×10^{-3})****
β_4 (Day ²)	3.1×10^{-7} (3.6×10^{-7})	5.7×10^{-5} (7.5×10^{-6})****

Note. For predictive modelling purposes, parameters with p values <0.10 and overall models with p -values of <0.10 were included.

^aNot included in final regression model because coefficient was found to be insignificant ($p > 0.1$) when simplifying the model structure.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

**** $p < 0.001$.

during summer and the contrasting timing for uptake and losses from the algal pool (Ford & Fox, 2014). Uptake was three times higher in summer ($0.154 \text{ gN m}^{-2} \text{ day}^{-1}$ and $0.021 \text{ gP m}^{-2} \text{ day}^{-1}$) as compared with the next highest season, spring ($0.053 \text{ gN m}^{-2} \text{ day}^{-1}$ and $0.007 \text{ gP m}^{-2} \text{ day}^{-1}$). Uptake was at a maximum in July and minimum in December. Biochemical losses were also highest in summer ($0.110 \text{ gN m}^{-2} \text{ day}^{-1}$ and $0.015 \text{ gP m}^{-2} \text{ day}^{-1}$); however, contrasting

uptake, peak biochemical losses occurred in September. Scouring of algal biomass was steadier throughout the year, but still was a maximum in summer with peak values occurring in July.

Results of the baseline loading scenario (i.e., regression model results) show that ratios of Sp_{av} to $ST1$ concentrations are slightly greater than one in winter (reflecting either a systematic bias due to arithmetic averaging or an additional diluted flow source), but the largest deviations are observed in spring and summer, coinciding with peak values of algal transformations. During nonsummer months, Sp_{av} concentrations were found to be higher than $ST1$ for NO_3^- (10–30%) and DRP (1–8%). One potential explanation for this is that, per our method, we used arithmetic as opposed to flow-weighted averages (in absence of flow measurements at all springs) that may bias the results either towards high or low nutrient concentrations. In our case, we surmise this would be attributed to biasing towards higher concentrations from smaller springs. A second explanation is that additional flow sources (e.g., wet weather seeps, lateral subsurface flow, or overland flow) that are less concentrated in NO_3^- and DRP at peak flows become significant contributors. Regardless, the ratios of both NO_3^- and DRP were relatively steady in fall through spring. Regarding the summer ratios (July–September), concentrations at Sp_{av} were 60–100% greater than $ST1$ for NO_3^- and 18–25% greater for DRP , highlighting major deviations.

To determine the ability of the algae fate and transport scenarios to explain deviations, we ran the scenario analysis for algal uptake and biochemical losses, in which results were able to explain deviations for DRP but only partially explain for NO_3^- (Figures 7, 8). For DRP , Scenario 1 resulted in concentrations being approximately 20% lower for Sp_{av} as compared with $ST1$, whereas Scenarios 2 and 3 resulted in concentrations 4–16% higher for Sp_{av} as compared with $ST1$. Therefore, considering the fate of algae allowed us to bound the 1–8% increase range observed throughout the remainder of the

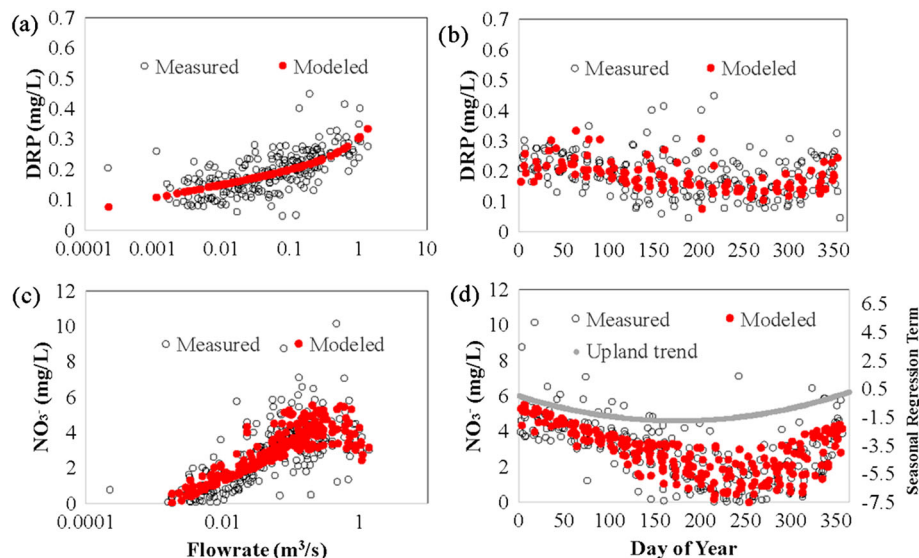


FIGURE 4 Comparison of measured versus predicted nutrient concentrations based on regression modelling in Table 3 for dissolved reactive phosphorus (DRP) and NO_3^- at the watershed outlet ($ST1$). Results of the model are plotted as a function of presumed important variables (flowrate and day of year). The upland trend in Figure 4d is provided to emphasize importance of both flow and upland seasonal processes to influence stream nitrate concentrations

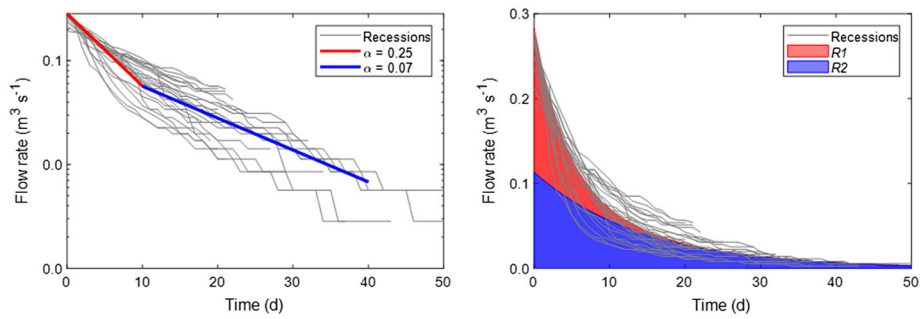


FIGURE 5 Master recession curve constructed from 32 recessions over 10 years of flow at ST1. The master recession curve was decomposed into two linear drainage reservoirs: R1 ($\alpha = 0.25$) and R2 ($\alpha = 0.07$). The area shaded in red is associated with quick and intermediate flow whereas the area shaded in blue is associated with slow flow

TABLE 4 Seasonal breakdown of hydrologic and nutrient fluxes and pathways for the Camden Creek Watershed

Season	Q_{ST1} ($\text{m}^3 \text{s}^{-1}$)	Q_{R1} ($\text{m}^3 \text{s}^{-1}$)	Q_{R2} ($\text{m}^3 \text{s}^{-1}$)	f_{Q-R1}	f_{Q-R2}
Winter (Jan–Mar)	0.30	0.07	0.23	24%	76%
Spring (Apr–Jun)	0.20	0.04	0.16	21%	79%
Summer (Jul–Sep)	0.06	0.02	0.04	34%	66%
Fall (Oct–Dec)	0.15	0.04	0.11	26%	74%
Season	$\text{NO}_{3\text{ST1}}$ (kgN day^{-1})	$\text{NO}_{3\text{R1}}$ (kgN day^{-1})	$\text{NO}_{3\text{R2}}$ (kgN day^{-1})	$f_{\text{NO}_3\text{-R1}}$	$f_{\text{NO}_3\text{-R2}}$
Winter (Jan–Mar)	94.95	18.63	76.32	20%	80%
Spring (Apr–Jun)	43.48	5.61	37.87	13%	87%
Summer (Jul–Sep)	7.80	1.88	5.92	24%	76%
Fall (Oct–Dec)	55.74	12.44	43.30	22%	78%
Season	DRP_{ST1} (kgP day^{-1})	DRP_{R1} (kgP day^{-1})	DRP_{R2} (kgP day^{-1})	$f_{\text{DRP-R1}}$	$f_{\text{DRP-R2}}$
Winter (Jan–Mar)	5.92	1.69	4.23	29%	71%
Spring (Apr–Jun)	4.61	1.37	3.24	30%	70%
Summer (Jul–Sep)	0.71	0.20	0.51	28%	72%
Fall (Oct–Dec)	3.97	1.18	2.78	30%	70%

year. Regarding NO_3^- , Scenarios 1 and 2 reflected the most extreme conditions and still resulted in concentrations that were 43–65% higher for Sp_{av} as compared with ST1. Results from Scenario 3 (which was most realistic for DRP) resulted in NO_3^- concentrations upwards of 80% higher for Sp_{av} as compared with ST1. This result suggests alternative removal mechanisms in the stream channel or adjacent riparian corridor for N dynamics that cannot be accounted for by algal biomass dynamics alone and will be highlighted in the discussion section.

4 | DISCUSSION

4.1 | Upland drainage controls on nutrients in karst agroecosystems

Hydrology of the Camden Creek watershed can be characterized by a prominent slow flow drainage reservoir and a second reservoir

reflecting a mixture of quick and intermediate flow paths. Our immature karst watershed had two distinct log-linear regions in the master recession curve (Figure 5), suggesting drainage of two distinguishable hydrologic reservoirs. For context, we compare with a mature karst system 21 km from our study watershed (Royal Spring basin). For Royal Spring, three distinguishable hydrologic reservoirs were identified in the master recession curve, with coefficients of 0.50, 0.15, and 0.05 day^{-1} reflecting the quick, intermediate, and slow flow paths, respectively (Husic, Fox, Ford, et al., 2019). Regarding slow flow, our results found a master recession coefficient of 0.07 day^{-1} suggesting 75% of flow in the watershed on an annual basis is governed by recharge through low permeability matrix pores and small fissures. The inverse of the recession coefficient ($1/\alpha$) represents the time that it would take to completely drain a reservoir assuming no additional recharge or recession slope changes (Tobin & Schwartz, 2016), and is calculated as 14 days for slow flow in Camden Creek. This value is short relative to slow flow drainage in Royal Spring (20 days). This is likely attributed to the immature, vadose zone conduit architecture

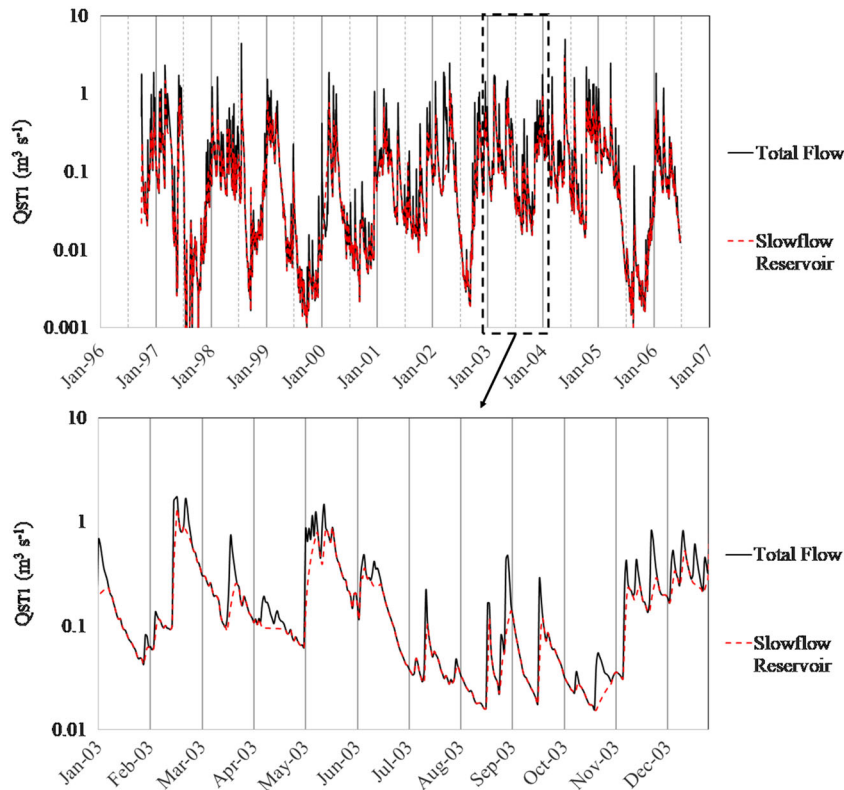


FIGURE 6 Results of the continuous flow recession analysis for the 10-year study at ST1. Results are presented for the total monitored flow and for the slow flow reservoir. Quick and intermediate flow is the difference between total and slow flow. An enhanced image of 1 year of recession analysis is provided to highlight quick/intermediate and slow flow variability within events for January 2003–December 2003

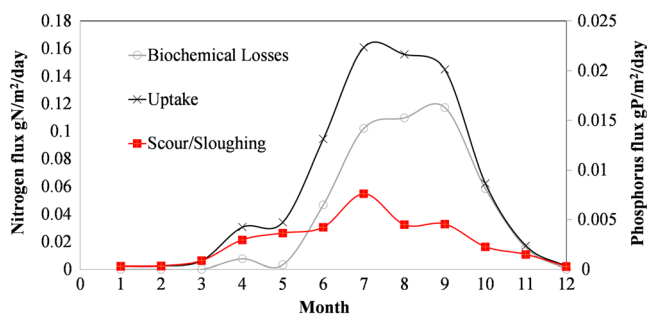


FIGURE 7 Results from a 5-year modelling study of algal C dynamics (Ford & Fox, 2014) were used to estimate algal N and P dynamics by considering analogous stoichiometry of processes and a C:N:P ratio of 40:7.2:1 (Chapra et al., 2014). Values from Ford and Fox (2014) were averaged for month of year (with January representing Month 1 and December representing Month 12). Uptake of N and P, losses associated with biochemical processes, and physical scouring/sloughing of algae are considered (note that uptake is equal to scouring plus biochemical losses on an annual basis)

of the Camden Creek springs that are, on average, about 15 m below the ground surface and drain small groundwater basins on the order of 1 km² (Keagy et al., 1993). It is important to note that these drainage values do not consider temporary storage within the aquifer reservoirs (e.g., field capacity, capillary forces, aquitards, and spatial heterogeneity in permeability). For example, in Royal Spring, numerical modelling indicated a mean residence time of 122(±9) days (Husic, Fox, Ford, et al., 2019), which is much greater than the aforementioned 20-day drainage time. Regarding the second hydrologic reservoir (R2), we

found a master recession coefficient of 0.25 day⁻¹, corresponding to a drainage time of roughly 4 days, representing a mixture of quick and intermediate flow pathways. This pathway likely reflects surface run-off that is redirected to sinkholes and perched aquifer drainage through large epikarst fractures and conduits that has been observed to occur above the bedrock surface (Keagy et al., 1993; Mellander et al., 2013). Indistinguishability of quick and intermediate pathways reflects the lack of resurgent streams, which provide the immediate spring response in mature karst systems. The implication of this finding is that hydrograph separation techniques may have limited applicability in less mature karst landscapes for differentiating soil hydrologic processes (e.g., piping through macropores) from tertiary pathways such as sinkholes and swallets. Findings from the Camden Creek analysis agree with the prevailing thought that the epikarst acts as a dynamic zone of water transfer and storage during moderate to extreme events (Aquilina, Ladouche, & Dörfli, 2006; Champollion et al., 2018; Husic, Fox, Adams, et al., 2019; Williams, 2008).

Synthesis of our NO₃⁻ pathway loading analysis with the results of regional karst watersheds suggests that soil water N dynamics regulate the annual NO₃⁻ load and are responsible for the low variability in NO₃⁻ loading across Inner-Bluegrass karst systems with varying degrees of fluviokarst development. We find that slow flow is the predominant contributor to watershed nitrate loads in the Camden Creek watershed due to higher volumetric flow contributions combined with higher nutrient concentrations, highlighting the importance of subsurface NO₃⁻ dynamics to regulate watershed NO₃⁻ fluxes in the immature karst system. Our finding of diluted NO₃⁻ concentrations during

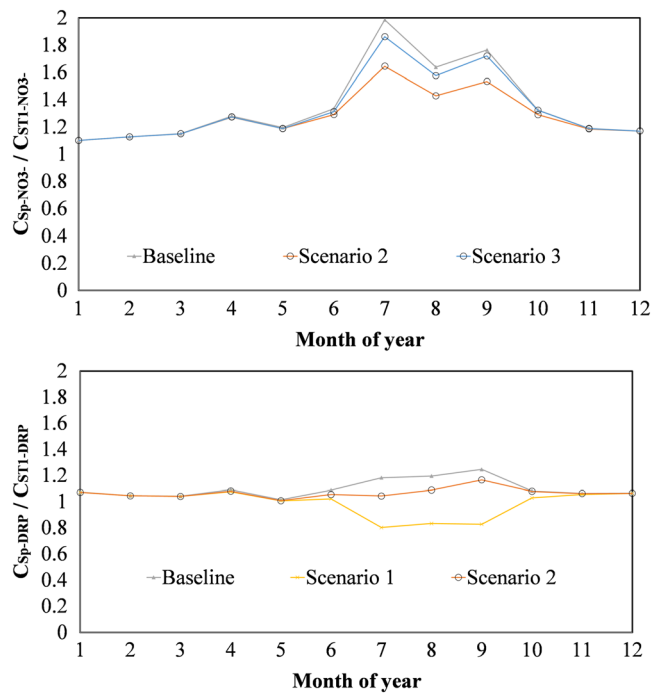


FIGURE 8 Results from the algae scenario analysis to illustrate differences in the ratio of flow weighted mean concentrations from Sp_{av} and the watershed outlet $ST1$. Results are provided for NO_3^- and dissolved reactive phosphorus (DRP). Scenario 1 assumed all algae biomass losses were lost as dissolved organic matter (i.e., dissolved organic nitrogen and dissolved organic phosphorus). Scenario 2 assumed that all algae losses were a result of mineralization/respiration to inorganic forms and that no nitrification occurred (i.e., total ammoniacal-N [TAN] and dissolved reactive phosphorus [DRP]). Scenario 3 assumed that all algae losses were a result of mineralization and that all TAN was subsequently nitrified (i.e., NO_3^- and DRP)

peak flows, as evidenced by the decreasing linear trend in Figure 4, is reflective of NO_3^- dynamics in many karst N agroecosystem studies (e.g., Husic, Fox, Ford, et al., 2019) and reflects rapid connectivity of nitrate-depleted precipitation mixing with the vadose zone NO_3^- source. Further, seasonality in NO_3^- concentrations was found to be a maximum in winter and minimum in summer and has been previously postulated to be reflective of physical and biochemical transformations in soil, epikarst, and phreatic zones broadly across agroecosystems (Exner-Kittridge et al., 2016; Ford, Williams, & King, 2018; Griffiths et al., 2012; Mulholland et al., 2008; Peterson et al., 2001; Pionke et al., 1999; Royer, Tank, & David, 2004). Our watershed loading analysis suggested an average annual loading of $1.7 \text{ tN km}^{-2} \text{ yr}^{-1}$. This result is similar to average annual loadings from both more mature ($1.8 \text{ tN km}^{-2} \text{ yr}^{-1}$) and less mature ($2.1 \text{ tN km}^{-2} \text{ yr}^{-1}$) karst agroecosystems previously reported in the study region, which have similar soil types and agroecosystem management (Ford et al., 2017; Husic, Fox, Ford, et al., 2019; Mahoney et al., 2018). These loadings have particularly low variability when considering the ranges reported in other disturbed karst watersheds; for instance, $4.70 \text{ tN km}^{-2} \text{ yr}^{-1}$ was observed from a 97-ha karst dairy farm spring in Ireland (Fenton et al., 2017), $10.38 \text{ tN km}^{-2} \text{ yr}^{-1}$ from the karstified Krathis River in

Greece (Tzoraki & Nikolaidis, 2007), and $7.27 \text{ tN km}^{-2} \text{ yr}^{-1}$ from a Long Term Ecological Research karst site in Austria (Hartmann et al., 2016). The implication of these findings is that physical and biochemical processes in the soil (coupled with upland management practices) may be the most important control on annual NO_3^- loading as compared with epikarst and phreatic zone dynamics. Future studies should explicitly consider the role of soil physical and biochemical properties as well as nutrient management dynamics when studying nutrient flow pathways in subsurface drainage (e.g., Williams, King, Ford, Buda, & Kennedy, 2016). Recent work has highlighted the ability of reservoir models in karst landscapes to separate the influence of soil and epikarst regions on vadose zone N processing (Husic, Fox, Ford, et al., 2019) and may be valuable tool in these future studies. Our findings have implications for targeting optimization of NO_3^- removal in soil vadose zones for receiving waters where long-term annual nitrate fluxes are of particular concern to the downstream waterbody. Given that timing of these loadings within the year can be highly variable as a function of karst maturity (Husic, Fox, Ford, et al., 2019), management practices may therefore have varying degrees of effectiveness at reducing annual nitrate loads and the timing of fluxes at finer time-scales also require consideration.

Findings from our study suggest that DRP loading is dominated by slow flow despite higher concentrations in quick and intermediate flow, which agrees with dynamics observed in similar karst agroecosystems; however, we find concentrations (and subsequently fluxes) of DRP to be an order of magnitude higher than other karst agroecosystem studies (e.g., Mellander et al., 2013), reflecting the limited capacity of the phosphatic limestone in our study site to attenuate DRP. We found DRP concentrations in Sp_{av} had a significant linear increase with increasing flow, suggesting that quickflow pathways were more highly concentrated in DRP. This finding agrees with existing theory that increasing connectivity of subsurface pathways to P-rich surface soils via macropores and preferential pathways will bypass the filtration capacity of soil and epikarst matrix, resulting in elevated DRP concentrations (Baker et al., 2017; Jarvie et al., 2014; King et al., 2014; Mellander et al., 2013). Nevertheless, slow flow concentrations were also high in our system with average DRP concentrations of 0.233 mg L^{-1} , a value that is nearly an order of magnitude higher than eutrophic thresholds of $0.02\text{--}0.03 \text{ mg L}^{-1}$ for freshwater algal proliferation (Table 4; Dodds, Smith, & Lohman, 2002; Sharpley et al., 2003). Mellander et al. (2013) collected high resolution data from a 92% permanent pasture karst watershed with mean Mehlich 3P of 66 ppm in the upper 10 cm of the soil profile and found mean TRP (unfiltered molybdate reactive P) concentrations to average 0.018 mg L^{-1} over a 2-year period, postulating that P attenuation may occur in all pathways due to transient sediment storage in dissolution conduits and fissures. Jarvie et al. (2014) similarly found that karst flow pathways may provide a significant sink (as much as 96% of SRP) during stormflows resulting in concentrations an order of magnitude lower than edge-of-field surface run-off. On the basis of the existing perception that karst networks attenuate P in soils and conduit sediments, our findings would suggest that sediments and soils of the Inner-Bluegrass Region may have limited capacity in both the

soil matrix and transient sediment stores in fractures to retain P once leached from surface soils.

4.2 | Nutrient fate in karst bedrock streambeds

Findings from this study suggest that algal biomass dynamics in the bedrock-controlled streams could govern seasonal production of TAN and TOC, as well as seasonal transient removal of DRP from the water column. Our results showed that timing of peak biochemical losses from the algae pool corresponded to maxima in the significant seasonal fluctuations of both TOC and TAN (Figure 3; Figure 7) suggesting the importance of algal leaching and mineralization to govern organic carbon and TAN dynamics in the stream channel. The significance of DOC leaching in warm summer months is recognized in that it provides a labile C source for heterotrophic bacteria, such as denitrifying bacteria, to attenuate contaminants (Arango & Tank, 2008; Ford et al., 2017; Hotchkiss & Hall, 2015; Lane, Lyon, & Ziegler, 2013; Marcarelli, Baxter, Mineau, & Hall, 2011). Conversely, pulses of TAN have been shown to enhance toxicity of harmful algae blooms and production of microcystin in laboratory studies and thus may become problematic in downstream waterbodies (Davis, Bullerjahn, Tuttle, McKay, & Watson, 2015). Regarding DRP attenuation, results in Figure 8 for the hypothetical algae fate scenario suggests that a significant fraction of DRP is transiently stored (and ultimately lost) to downstream waterbodies as particulate algal P due to sloughing and scour. If benthic algal biomass dynamics from the nearby watershed are congruent with our system, an average of $0.07 \text{ kgP day}^{-1}$ is transported during summer (equivalent P concentration of 0.015 mgP L^{-1}) and would be susceptible to regeneration downstream. As previously mentioned, given eutrophic conditions of $0.02\text{--}0.03 \text{ mgP L}^{-1}$, assessing the fate of this flux is significant for borderline eutrophic agroecosystems.

We were unable to fully describe NO_3^- summer deviations in the Camden Creek watershed from algal biomass dynamics alone, leading us to postulate alternative biotic controls in the bedrock-controlled streambed. As highlighted in the results, algal biomass dynamics did not fully explain nitrogen removal dynamics in the stream channel. In order to attain the concentration ratio of $S_{p_{av}}$ to $ST1$ from fall, winter, and spring months (average ratio of 1.19), an average of 2.5 kgN day^{-1} ($0.2 \text{ gN m}^{-2} \text{ day}^{-1}$) would need to be regenerated to the stream channel, suggesting the stream has a potentially large nitrate sink that is not able to be explained by algal biomass dynamics. Of interest to watershed managers and scientists is whether such a NO_3^- sink is associated with assimilatory transient storage in other biota that is subsequently exported from the watershed or from dissimilatory processes (i.e., denitrification) that would permanently remove NO_3^- from the waterbody. Given the findings of the algal scenario analysis for DRP, we postulate that permanent removal via denitrification is the most likely contributor. Although riparian zone denitrification may contribute, we do not anticipate it to be the governing mechanism based on previous research that has shown the lowest rates of denitrification in riparian zones occur at the soil-stream interface (Ranalli &

Macalady, 2010). Although studies have shown that algal mats provide little influence on overall stream denitrification in other midwestern agroecosystems (e.g., Schaller, Royer, David, & Tank, 2004), we postulate proliferation of thick mats of floating aquatic macrophytes may facilitate high rates of N removal in the bedrock stream.

Proliferation of floating aquatic macrophytes, commonly observed at the study site, likely reflect the favourable conditions created by mineralization of benthic algae and the limited nitrification potential of systems flowing over bedrock. Our results support peak TAN concentrations in late summer to occur at stream sites and found limited variability in $S_{p_{av}}$, suggesting that in-stream production of TAN is the primary source. Although benthic substrates have been recognized to facilitate rapid nitrification of TAN within tens to hundreds of metres of being introduced to the stream reach (Peterson et al., 2001), the lower metabolic activity of bedrock streambeds (Argerich et al., 2011) likely leads to lower rates of nitrification. The rapid growth of floating aquatic vegetation, namely duckweed, is widely recognized in low-velocity freshwater systems where TAN is present, thus outcompeting and shading algal biomass (Peng, Wang, Song, & Yuan, 2007; Ran, Agami, & Oron, 2008). Given the timing of high TAN concentrations in the stream channel to occur during low-flow and long-residence periods, our findings suggest that regenerated TAN may subsequently fuel rapid floating aquatic macrophyte growth. Although we did not have quantitative measures of floating aquatic macrophytes, we identified duckweed as the prevalent floating macrophyte and observed thick mats to occur during low-flow periods from late spring to early fall. Aerial imagery from September highlights the prevalence of aquatic vegetation in the stream channel during summer (Figure 1). Such dynamics have not previously been highlighted as a major control on N dynamics in other agroecosystem streams to our knowledge and need further validation.

We postulate that the floating aquatic vegetation provides a substrate for microbial biofilms and, when coupled with a labile DOC source, may facilitate denitrification rates that are on the high side of disturbed agroecosystem streams. Our findings show that leaching of algal C via organic carbon exudates is greatest during the period of maximum NO_3^- depletion. Denitrifying bacteria are heterotrophic organisms and will remove NO_3^- more effectively in environments enriched in labile C, such as autochthonous biomass exudates (Lane et al., 2013). Further, large mats of aquatic macrophytes have been found to promote denitrification on plant surfaces by supporting microbial biofilms (Eriksson, 2001; Eriksson & Weisner, 1999; Veraart, de Bruijne, de Klein, Peeters, & Scheffer, 2011). Assuming that, conservatively, 10% of the aforementioned nitrate sink can be explained by denitrification (Mulholland et al., 2008), a denitrification rate of $0.02 \text{ gN m}^{-2} \text{ day}^{-1}$ or $8.3 \times 10^2 \text{ } \mu\text{gN m}^{-2} \text{ hr}^{-1}$ during peak removal is plausible. This estimate would place the bedrock stream in the upper 50th percentile of denitrification rates reported in agricultural streams (Mulholland et al., 2008). Although this hypothetical estimate needs further quantitative validation using more sophisticated modelling and field monitoring approaches that consider the complex interactions of N species within the various biotic pools, these findings provide evidence of the significance of this spring-surface water

interface to be a hotspot for biogeochemical transformations of N and provides testable hypothesis for future work.

5 | CONCLUSIONS

The findings from our study highlight congruencies and contradictions to prevailing karst nutrient dynamics at the watershed scale. Hydrology of the immature karst watershed displayed two distinguishable reservoirs in which slow flow reflected 75% of discharge and quick/intermediate flow represented 25%. This finding contrasted a nearby mature karst watershed in which tertiary porosity allowed differentiation of quick and intermediate flow paths. Nitrate dynamics in uplands were congruent with seasonal dynamics in other karst and nonkarst agroecosystems; however, in-stream processes deviated and highlight the potential importance of denitrification and floating aquatic vegetation to influence NO_3^- loadings, particularly in summer. Regarding P, we found DRP to be heightened in slow flow pathways due to reduced retention capacity of the soils that are derived from Ordovician limestone, which conflicts other karst agroecosystem studies; however, in-stream dynamics were well-constrained by perceptions of algal fate. Although the long-term ambient dataset allowed us to differentiate significant seasonal trends and perform regression analysis, we highlight the need for integration of new innovations including numerical models and cutting-edge monitoring technologies to continue the evolution of our knowledge of karst nutrient dynamics. On the basis of our results, we identify several areas for future work, including the following:

1. Identifying and testing methods for differentiation of quick and intermediate flow pathways of contaminants in immature karst.
2. Deterministic modelling of in-stream fate of nitrate as influenced by competing biotic pools (e.g., algae, floating aquatic plants, and microbial communities).
3. Differentiating anthropogenic and allogenic sources of DRP for soil derived from phosphatic lithology.

Ultimately, constraint of these processes and development of predictive tools will enable improved management of nutrients in complex karst agroecosystem watersheds.

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